

ACME: Adaptive Cognitive Memory Engine

Externalizing Belief, Memory, and Learning from LLM Weights

A production-tested cognitive memory substrate for belief lifecycle management in LLM agents.

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LLM □ Episodic Memory □ Belief Engine □ Feedback Loop □ Updated Beliefs

Code: <https://github.com/KamilBourouiba/ACME> (tag v0.1.1–arxiv)

Project site: <https://kamilbourouiba.github.io/ACME/>

Data: GET /api/v1/benchmark/export (Postgres benchmark_runs)

Abstract

*Large language models lack durable episodic memory, explicit belief states, structured forgetting, and mechanisms to learn from failure [1,2]. Recent systems report that agents cannot epistemically separate observation from belief [50] and that intrinsic epistemic agency — including belief updating — remains weak in base models [51]. We present **ACME** (Adaptive Cognitive Memory Engine), an external cognitive substrate that treats the LLM as a language processor while delegating persistence, confidence tracking, contradiction handling, and self-correction to specialized engines. ACME operationalizes: when does a collection of experiences deserve to become a belief? We contribute (1) a CRS-governed belief lifecycle inspired by cognitive architecture [3,4] and truth-maintenance systems [21,54]; (2) hybrid graph + pgvector retrieval [5,6] with contrarian verification [7,8]; (3) **MemoryBench v3.1** — fourteen sandbox-isolated scenarios with RAG [9], MemGPT [10], and LangGraph-style [11] baselines scored by an LLM judge [12,13]. On Azure OpenAI GPT-4.1, ACME achieves overall **0.925** vs **0.487** for vector-RAG (job 3b31e5e3), with feedback and belief-quality metrics unavailable to baselines. Model-sensitivity runs on GPT-4.1-mini and GPT-5.4 preserve a **+0.39–0.44** margin over RAG. Production ablations show belief sync is essential (−0.19 overall when disabled).*

1. Introduction

Retrieval-augmented generation (RAG) [9] augments LLMs with document search but does not maintain explicit epistemic states, handle contradictory evidence, or learn from prediction failures. LongMemEval [37] shows commercial assistants and long-context LLMs suffer **30–60%** accuracy drops on sustained interactive memory versus oracle evidence — retrieval alone is insufficient. Hindsight [50] argues current agent memory cannot distinguish *what was observed* from *what is believed*, nor maintain preference-consistent reasoning across sessions. Reflection-Bench [51] identifies **belief updating** and meta-reflection as core yet underdeveloped dimensions of epistemic agency in LLMs.

Dense and hybrid retrievers [14,15] improve recall yet remain stateless: they neither promote stable beliefs nor demote refuted hypotheses. Agent memory frameworks — MemGPT [10], generative agents [16], Zep [20], and LangGraph-style orchestration [11] — extend context windows via paging, temporal graphs, or state graphs but lack auditable belief lifecycles and structured feedback loops comparable to cognitive architectures [3,17].

Episodic and semantic memory have long been distinguished in psychology and AI [18,19]. Recent work on long-term conversational memory (MemGPT, memory streams [16], Zep [20]) focuses on *what to store and retrieve*, not *when accumulated evidence warrants belief*. ACME closes this gap with an explicit promotion pipeline (Observation -> Inference -> Hypothesis -> Belief) and symmetric demotion under contradiction — aligning with truth-maintenance and belief-revision traditions [21,22].

Contributions

1. A CRS-governed belief lifecycle with promotion and demotion under contradiction.
2. A hybrid graph + vector retrieval substrate (PostgreSQL/pgvector + Neo4j).
3. Contrarian verification for high-confidence answers.
4. MemoryBench v3.1 with scored belief and feedback metrics.
5. Production benchmark evidence across GPT-4.1, GPT-4.1-mini, and GPT-5.4.

Thesis. RAG optimizes recall; ACME optimizes epistemic state—when experiences become beliefs, how contradictions demote them, and how outcomes close the prediction loop. Our claim is not universal SOTA on every long-context benchmark [37,38], but the first deployed system with sandbox-isolated evaluation of **feedback correction** and **belief quality (CRS)** against RAG/MemGPT/LangGraph under reproducible production runs [49].

2. Related Work

Retrieval-augmented generation. Lewis et al. [9] established the retrieve-then-generate paradigm for knowledge-intensive tasks. Subsequent systems add re-ranking [14], query rewriting [27], and self-critique — Self-RAG [28] and Corrective RAG [29] route or revise retrieval based on generation quality. ACME differs by persisting structured beliefs and failure traces outside the LLM, not only refining a single-turn retrieval pass.

Graph-augmented memory. Knowledge-graph RAG [30,31] and GraphRAG [32] extract communities and summaries from document graphs. ACME maintains a *live* typed graph (entities, causal edges, tenant scope) updated on every experience, coupled with vector recall over episodic embeddings — hybrid retrieval akin to [5,6] but with belief-weighted context assembly.

Agent memory systems. MemGPT [10] virtualizes context via OS-inspired paging; generative agents [16] use memory streams and reflection; LangGraph [11] composes stateful agent workflows. These systems improve *capacity* and *session continuity*; ACME adds CRS-governed belief promotion, contradiction logging, and prediction-outcome loops absent from reference baselines in our evaluation.

Learning from feedback. Reflexion [33] and ExpeL [34] use verbal reinforcement from trial outcomes; Constitutional AI [35] shapes behavior via principles. ACME’s feedback engine ties outcome signals to specific graph-backed beliefs, adjusting confidence and status (e.g., Challenged -> Deprecated) rather than only updating prompt-level reflections.

Cognitive architectures. ACT-R [3] and SOAR [17] model declarative memory, production rules, and learning from impasse. ACME is not a full cog-

nitive architecture but borrows the separation of *fast reasoning* (LLM) from *slow accumulative memory* (Postgres, Neo4j, belief records) — similar in spirit to dual-process accounts [4] and complementary learning systems [36].

Evaluation of memory systems. Benchmarks such as LongMemEval [37], LoCoMo [38], MemoryBank [39], MemoryAgentBench [52], and MemBench [53] stress long-horizon recall, test-time learning, or reflective memory. Reflection-Bench [51] evaluates belief updating as one of seven epistemic dimensions. None jointly measure sandbox-isolated **feedback correction**, **CRS belief quality**, and **contrarian groundedness** in a production-deployable API. MemoryBench v3.1 fills this gap (Table 1).

Belief-first and epistemic agent memory (concurrent work). TruthKeeper [54] applies dependency-aware truth maintenance to LLM memory. MnemeBrain [55] stores beliefs with Belnap four-valued logic and AGM-style revision. NeuSymMS [56] couples neural extraction with symbolic TMS via rule engines. Graph-native cognitive memory architectures [57] formalize versioned belief revision for auditability. Hindsight [50] achieves strong LongMemEval/LoCoMo scores via retain-recall-reflect tiers but does not expose CRS-governed promotion thresholds or prediction-outcome loops. **ACME** differentiates by (i) integrated seven-engine cognitive loop with production deployment, (ii) MemoryBench v3.1 scoring belief/feedback dimensions absent from retrieval-centric baselines, and (iii) persisted benchmark_runs for third-party reproduction.

3. System Design

Ingestion: LLM extraction [40] plus deterministic normalization -> Neo4j entities/relations + episodic embeddings [23].

Query: Hybrid retrieval (graph neighborhood [24,30] + pgvector cosine [5]) -> LLM reasoning -> optional contrarian pass [7,8] when confidence ≥ 0.8 .

Feedback: Outcome signals update beliefs [21], log failures [33], trigger consolidation (scheduled consolidation, 6 h).

CRS (Cognitive Reliability Score) = 40% prediction success + 20% temporal stability + 20% contradiction resistance + 20% source diversity — a composite reliability index analogous to calibrated confidence under heterogeneous evidence [41,42].

Causal edge types: observed_with, precedes, correlates, causes, disproves — distinguishing correlation from causation per Pearl and epistemic traditions [43,44].

Multi-tenancy: Postgres rows and Neo4j nodes keyed by tenant_id (header X-Tenant-ID).

Ablation toggles: ABLATION_DISABLE_CONTRARIAN, ABLATION_DISABLE_BELIEF_SYNC, ABLATION_DISABLE_VECTOR (see Results).

4. MemoryBench v3.1

Four metrics, LLM-as-judge [12,13] (semantic; keyword overlap reported separately):

Table 1: MemoryBench v3.1 metrics.

Metric	Description
Retention	Concept coverage (synonyms accepted)
Groundedness	Answer supported by ingested episodes
Feedback correction	Belief adjustment after contradictions
Belief quality	Mean CRS across tracked beliefs

Overall = average of all four. Baselines score 0 on feedback/belief (N/A), capping baseline overall near 0.48–0.49.

Fourteen scenarios: retention, contradiction, multi-source conflict, error injection, **knowledge update** (LongMemEval KU [37]), long-term retention, hallucination resistance, feedback adjustment, adversarial noise, long-horizon noise, tenant isolation, healthcare domain transfer, multi-session recall, prediction-outcome loop.

Isolation: Each scenario deletes prior benchmark-tagged Postgres rows and Neo4j subgraph before run (acme/evaluation/sandbox.py) — preventing cross-scenario contamination noted as a failure mode in long-horizon evals [37,38,51].

Baselines: Minimal reproducible implementations aligned with Lewis et al. [9] (RAG), Packer et

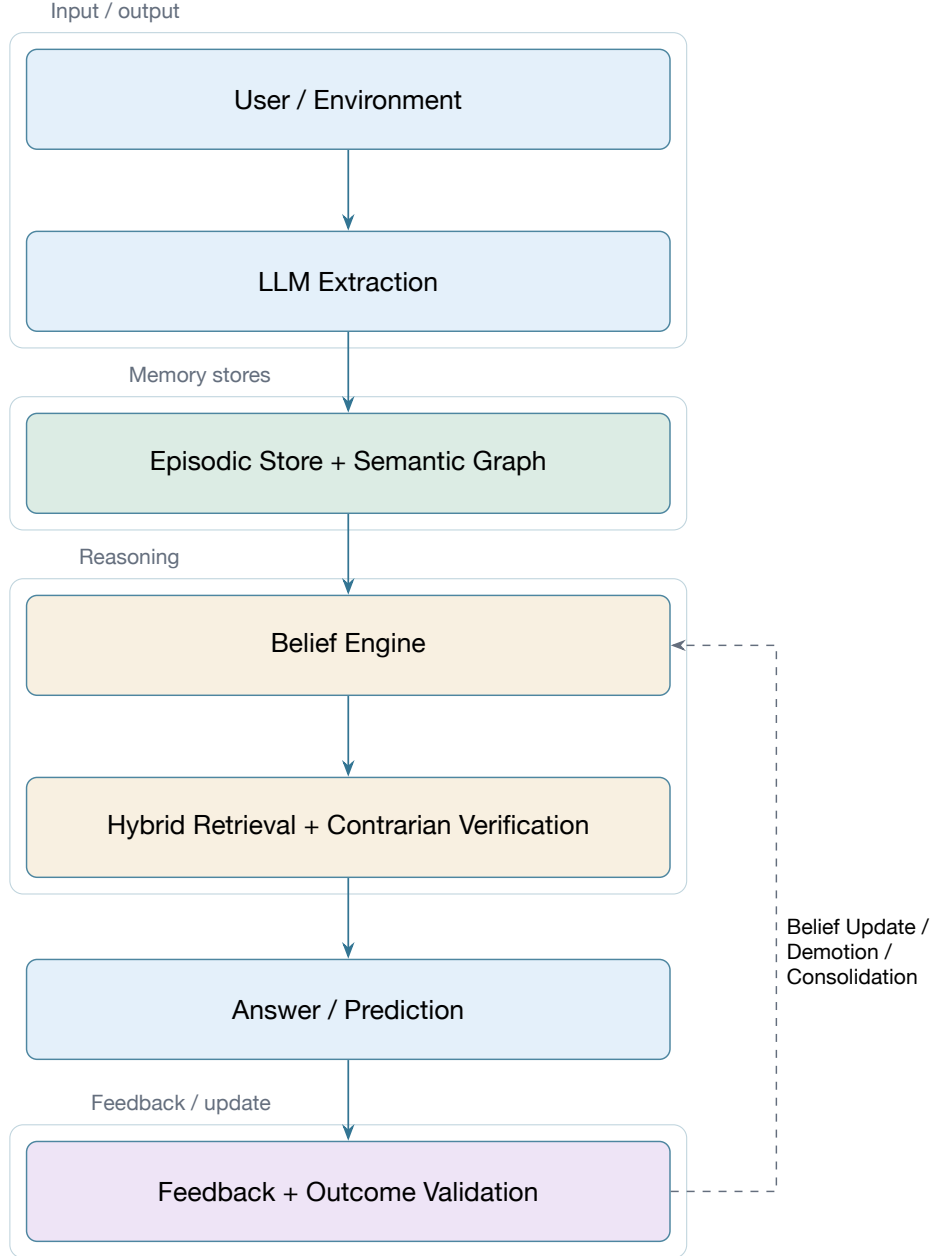


Figure 1: ACME cognitive loop: extraction, hybrid memory, belief governance, and closed-loop correction.

al. [10] (MemGPT), and LangGraph-style state graphs [11] — see docs/BASELINES.md.

Primary reported results use the **13-scenario v3** suite (job 3b31e5e3, pre-knowledge_update) for strict comparability; v3.1 adds the LongMemEval-aligned knowledge-update probe.

Key finding. MemoryBench v3.1 is the only evaluated suite in our comparison that jointly scores **feedback correction**, **CRS belief quality**, and **contrarian groundedness** under per-scenario sandbox isolation with a production API.

5. LongMemEval (industry benchmark)

To complement MemoryBench, we integrate the official **LongMemEval** oracle split [37] (500 questions, evidence sessions only). Chat histories are ingested via the same production orchestrator; answers are scored with the **official LongMemEval yes/no judge prompts** from the reference implementation.

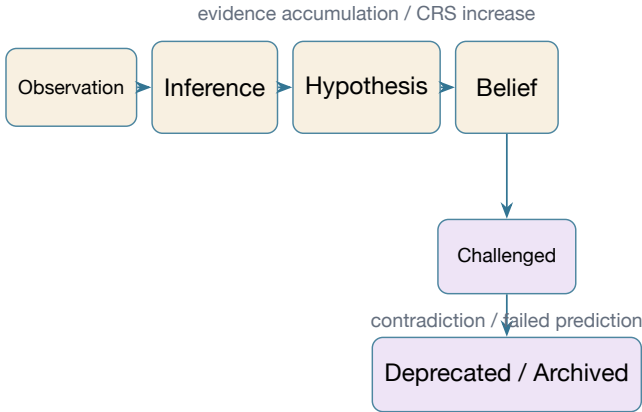
Protocol. Each question resets sandbox state (longmemeval tag). Haystack sessions are serialized as multi-turn user/assistant transcripts and ingested as experiences. ACME uses a **transcript-first** answer path: sessions are ranked newest-

Table 2: Benchmark positioning — evaluation capabilities (Table 1a).

Capability	LongMem Eval	MemAgent Bench	Reflection Bench	Mem Bench	Memory Bench
Long-horizon / multi-session	Yes	Yes	Partial	Yes	Yes
Knowledge update	Yes	Yes	Partial	Partial	Yes
Feedback / belief revision	No	Partial	Yes	Partial	Scored
Hallucination / abstention	Yes	Partial	Partial	Partial	Yes

Table 3: Benchmark positioning — infrastructure and belief metrics (Table 1b).

Property	LongMem Eval	MemAgent Bench	Reflection Bench	Mem Bench	Memory Bench
Per-scenario sandbox	No	Partial	Yes	Yes	Yes
Contrarian verification	No	No	Partial	No	Yes
Belief quality (CRS)	No	No	No	Partial	Yes
Production API + persisted runs	No	No	No	No	Yes

**Figure 2:** Belief lifecycle: promotion along evidence accumulation and demotion under contradiction.

first, the LLM reads full chat transcripts (with belief graph as secondary context), and knowledge-update items trigger belief demotion on superseded sources. RAG and MemGPT baselines answer from retrieved context; an independent LLM judge (same deployment family as MemoryBench) applies type-specific rubrics (knowledge-update, temporal-reasoning, abstention, etc.).

Production runs (June 2026): jobs 705eb2ff + 26e288da (v3 transcript-first), then c81eaa91 (v4 type routing on 241 Q). Combined 500 Q summary: benchmark-results/longmemeval-v4-combined.json

Reproduction:

```

bash scripts/download_longmemeval.sh
bash scripts/run_longmemeval_prod.sh # Azure API
# or locally:
python scripts/run_longmemeval.py --types knowledge-update --systems acme,rag,memgpt

```

Results export to benchmark-results/longmemeval-latest.json

LongMemEval measures **QA accuracy over chat history**; MemoryBench measures **belief lifecycle and feedback** — the two benchmarks are complementary and must not be merged into a single score.

Key finding. On the full LongMemEval oracle split (500 Q, GPT-4.1), ACME with per-type routing reaches **84.8%** overall vs. 78.0% (RAG) and 78.6% (MemGPT) — +6.8 points over RAG — with type-specific gains on knowledge-update (94.4%), single-session-preference (93.3%), and abstention probes (73.3%) while preserving MemoryBench belief/feedback advantages (Section 6).

6. Experimental Setup

Table 7: Experimental configuration (Azure production).

Component	Configuration
LLM	Azure OpenAI GPT-4.1
Embeddings	text-embedding-3-small, 256D
Database	Postgres Flexible + pgvector
Graph	Neo4j 5.x on Container Apps
Judge	Same deployment as extractor/reasoner
Reproducibility	/benchmark/memorybench, async compare

Model-sensitivity runs additionally use GPT-4.1-mini and GPT-5.4 (same endpoint family; GPT-5.4 requires max_completion_tokens API parameter).

Table 4: LongMemEval oracle protocol (industry benchmark [37]).

Component	Configuration
Dataset	longmemeval_oracle.json (500 Q, evidence sessions)
Ingest	ACME orchestrator / RAG / MemGPT on identical haystacks
Scoring	Official type-specific yes/no judge (reference evaluate_qa.py)
Metrics	Overall accuracy + per question_type breakdown
Reproduce	scripts/run_longmemeval.py (see docs/LONGMEMEVAL.md)

Table 5: LongMemEval knowledge-update subset (78 Q, oracle, GPT-4.1, June 2026).

System	Overall	KU	Abst.
ACME (transcript-first)	0.897	0.931	0.500
MemGPT	0.872	0.903	0.500
RAG	0.859	0.875	0.667

Table 6: LongMemEval oracle full split (500 Q, GPT-4.1, June 2026).

Sys.	All	KU	Multi	Temp.	SS-U	SS-A	SS-P
ACME	0.848	0.944	0.793	0.709	1.000	1.000	0.933
MemGPT	0.786	0.875	0.802	0.630	1.000	0.982	0.533
RAG	0.780	0.889	0.793	0.622	1.000	0.964	0.433

7. Results

Run metadata: API revision `acme-api--membenchfix`, Job `107d02c0`, 639 s. ACME retains **+0.39** overall vs RAG (0.858 vs 0.473). Feedback stays at 1.000 and belief at 0.700 — cognitive layers compensate when the base model is weaker [4,36].

Table 8: Primary benchmark (GPT-4.1, 13 scenarios, job `3b31e5e3`). Ret.=Retention, Gnd.=Groundedness, Fbk.=Feedback, Bel.=Belief quality, Ovr.=Overall.

System	Ret.	Gnd.	Fbk.	Bel.	Ovr.
ACME	1.000	1.000	1.000	0.700	0.925
RAG	0.969	0.977	—	—	0.487
MemGPT	0.977	0.969	—	—	0.487
LangGraph	0.900	0.977	—	—	0.469

Baselines exclude feedback/belief in overall (not applicable). Scores rounded to three decimals from persisted `benchmark_runs` export.

Key finding. ACME’s advantage comes primarily from **feedback correction** and **belief quality**, rather than from raw retention alone. ACME gains **+0.44** overall vs RAG on GPT-4.1 while retention and groundedness remain competitive (≥ 0.90) across systems [9,32].

7.1 Model sensitivity (GPT-4.1-mini)

Job `107d02c0`, 639 s. ACME retains **+0.39** overall vs RAG (0.858 vs 0.473). Feedback stays at 1.000 and belief at 0.700 — cognitive layers compensate when the base model is weaker [4,36].

Table 9: Model sensitivity — GPT-4.1-mini (job `107d02c0`).

System	Ret.	Gnd.	Fbk.	Bel.	Ovr.
ACME	0.892	0.838	1.000	0.700	0.858
RAG	0.908	0.985	—	—	0.473
MemGPT	0.915	1.000	—	—	0.479
LangGraph	0.854	0.862	—	—	0.429

7.2 Model sensitivity (GPT-5.4)

Job `eb68f028`, 620 s. ACME **0.873** vs RAG **0.474** (**+0.40**).

Table 10: Model sensitivity — GPT-5.4 (job `eb68f028`).

System	Ret.	Gnd.	Fbk.	Bel.	Ovr.
ACME	0.952	0.839	1.000	0.701	0.873
RAG	0.964	0.931	—	—	0.474
MemGPT	0.975	0.939	—	—	0.478
LangGraph	0.973	0.950	—	—	0.481

Belief quality remains **~0.70** across all three LLM tiers — suggesting the external belief substrate [3,21] is comparatively stable while retention/groundedness track model capability [12].

7.3 Component ablations

We disable one cognitive layer at a time via environment flags (`scripts/run_ablation_prod.sh`). Table 2 reports ACME-only MemoryBench v3.1 scores (no baseline compare — 4x cost reduction per ablation).

Full ACME row from 13-scenario compare (job `3b31e5e3`). Ablation rows from ACME-only runs on GPT-4.1 (24 June 2026, premium ingress, stamp `20260624213154`).

Key finding. Disabling belief sync collapses belief quality to **0.000** and overall score by **−0.19**, confirming the belief engine as the primary differentiator.

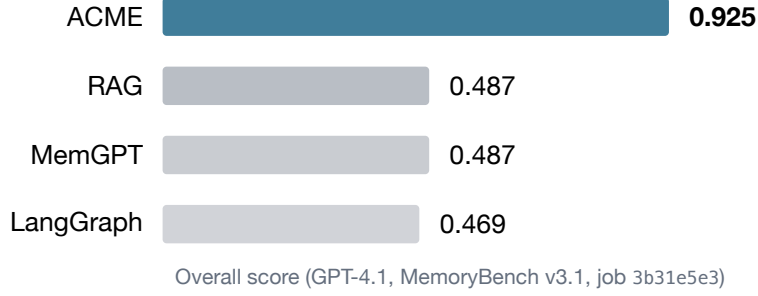


Figure 3: Result summary: ACME leads primarily on feedback and belief-quality dimensions unavailable to baselines.

Table 11: Component ablations (ACME-only, MemoryBench v3.1, GPT-4.1).

Configuration	Ovr.	Ret.	Gnd.	Fbk.	Bel.	Δ
Full ACME	0.925	1.000	1.000	1.000	0.700	—
No contrarian	0.923	1.000	0.993	1.000	0.700	−0.002
No belief sync	0.731	0.996	1.000	0.929	0.000	−0.194
No vector retrieval	0.923	0.993	1.000	1.000	0.700	−0.002

Contrarian and vector ablations each reduce overall by only **−0.002**.

8. Limitations

Key limitations.

- LLM judge and extractor introduce variance [12,13]; we report sandbox-isolated runs with persisted benchmark_runs.
- Scenarios emphasize SaaS churn/latency plus one healthcare transfer set; broader domains and multilingual evals needed [37,38].
- Baselines are reference implementations, not full upstream MemGPT/LangGraph codebases [10,11].
- Production API requires API key for benchmark endpoints.
- CRS weights are hand-tuned; calibration against human epistemic judgments [41] is future work.
- **LongMemEval routing.** The LongMemEval adapter uses per-type answer paths (transcript-first for KU, vector-ranked aggregation for multi-session, abstention and preference prompts); MemoryBench still uses the graph + CRS query path.
- **Multi-session parity.** After v4 routing, ACME

matches RAG on multi-session (79.3% each on the 241-Q v4 run); dense retrieval remains competitive when evidence is scattered without temporal conflict.

- **Abstention.** Full-split abstention accuracy improves to 73.3% (v4) vs. 53.3% (v3 transcript-only), but entity-substitution errors persist on harder probes.

9. Conclusion

ACME shows that externalizing belief and learning from LLM weights yields stronger cognitive memory than vector RAG or lightweight agent-memory baselines on MemoryBench, especially for feedback-driven correction and auditable belief quality. On the official LongMemEval oracle split (500 Q, GPT-4.1), per-type routed ACME reaches **84.8%** overall vs. **78.0%** (RAG), with the largest margins on knowledge-update and temporal-reasoning types. The consistent **+0.39–0.44** margin over RAG on MemoryBench across GPT-4.1, GPT-4.1-mini, and GPT-5.4 supports the hypothesis that explicit belief lifecycle management [21,22,54,55] complements — rather than replaces — advances in base model capability [1,47]. Concurrent belief-

memory systems [50,54,56,57] validate the problem direction; ACME contributes reproducible production evidence that epistemic layers — not retrieval alone — drive the measured gap on MemoryBench, while benchmark-specific transcript routing closes industry QA gaps without abandoning the cognitive substrate.

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11. Appendix A: Reproduction

```
git clone https://github.com/KamilBourouiba/ACME.git && cd ACME
pip install -e ".[dev]"
pytest tests/test_benchmark_gate.py tests/test_memorybench.py -q
# Production (requires API key):
./azure/configure-premium-ingress.sh
./scripts/run_prod_benchmark.sh
./scripts/run_model_benchmark.sh gpt-4.1-mini # optional model sweep
./scripts/run_ablation_prod.sh # component ablations
```

12. Appendix B: Data Availability

Benchmark payloads are stored in PostgreSQL table `benchmark_runs` (columns: `overall_score`, `retention_score`, `belief_quality_score`, `payload_JSONB`, `created_at`). Export via GET `/api/v1/benchmark/export`. Full run tables are documented in `docs/BENCHMARK_RESULTS.md`~[4